INSTITUTIONAL COURSE SYLLABUS SE411

Course Code: SE411	Course Title: Software Construction
Course Instructor: Dr. Asma Sattar	Email: asattar@psu.edu.sa
Credit Hours: 3 (3, 0, 1)	Course Location: Girls Campus
Scheduled Office Hours: 5	Monday, Wednesday: 9:00 AM – 10:00 AM, Sunday, Monday, Wednesday: 11:00 AM – 12:00 PM
Office Location: R305	
Co-Requisite: none	Prerequisite: SE401
On Campus or Online: On Campus (Tuesday sessions remote)	

Mission Statement

"The College of Computer and Information Sciences aims to offer internationally proven computing programs in an academic environment that promotes excellence and innovation in education, research and service to the community."

I. Course Description: This course will provide students with an in-depth study of software construction. Topics include basic theory of grammars and parsing, use of parser generators, software construction fundamentals (minimizing complexity, anticipating change, constructing for verification and standards in construction), managing construction (construction models, construction planning and construction measurement), practical considerations (construction design, construction languages, coding, construction testing, reuse, construction quality, configuration management, security, automation, and integration), and techniques for handling concurrency and inter-process communication.

II. Course Learning Outcomes: On the successful completion of this course, students will be able to demonstrate the following:

Skills	Course Learning Outcomes
Knowledge & Understanding	**CLO 1: Define and understand the fundamentals of software construction concepts** **CLO 3: Interpret key considerations in software**

	integration, testing, reuse, and quality
Skills	**CLO 2: Demonstrate the knowledge of the key software design, language, and coding concepts** **CLO 4: Evaluate the key construction concepts in terms of best practices** **CLO 5: Analyze and improve existing or new software projects using various software research techniques to ensure quality and innovation.**
Values	

III. Tentative Weekly Course Schedule: (Should mention the specific course topics to be covered within the semester) *May change to accommodate guest presenters & student needs*.

WEEK	UNIT/ TOPIC	CLO(s) alignment	STUDENT TASKS (assignments, readings, exams, quizzes, presentations, etc…)	Number of Contact hours
1	Introduction to software construction, Version Control with Git, unit testing. [Lab 01]	1	Labs, Quiz1, midterm and Project(part 1), final	4
2-3	Java generics and lambda expressions [Lab 02, 03,04]	1,4,5		8
4	Dependency management, Exception handling and logging [Lab 04, 05, 06]	1,4,5		4
5-7	Object-oriented design patterns implementation with Java. [lab 07, 08, 09, 10]	2,4,5		12
8	Concurrency with Java [lab 11, 12]	1,2,4,5		4
9-10	Java frameworks: Spring, Spring boot, inversion of control, Thymeleaf. [lab 13,14,15]	3,4,5	Labs, Quiz2, and Project(part 2), Lab exam, Final	8
11-13	Scalable distributed systems: Building REST API with Java, Message queues.	3,4,5		12
14-15	Application Security in Java [16,17,19]	3,4,5		8

IV. Student Assessment & Teaching Strategies:

Assessment Task (Indicate the kind of assessment tasks to be used to measure student learning in each of the learning domains. Example: quiz, oral examination, group work, etc.).

Teaching Strategies: (Please indicate the teaching and student activities to be used to develop the kinds of learning involved in each learning domain. Also, research specialized Information about Best Teaching Practices for the particular course/field).

Domain	Assessment Task	Teaching Strategy
Knowledge **CL01, CLO2, CLO3**	Quiz1, Midterm, project, lab exam, final	In class lectures, labs
Skills **CLO4** **CLO5**	Quiz2, Project, lab exam, final	Lectures, labs and project.
Values	none	

V. Course Requirements

[Whatever tasks and assignments you include in your course should be aligned with the specified learning outcomes (final learning, skills, knowledge, attitudes, and values the students leave the course with) you have defined and specified earlier.

These requirements should be consistent with the Course Specification on file in the particular department.]

VI. Schedule of Assessment

(Specify the schedule/date of assessment & proportion of assessment, attach related rubrics for each assessment if applicable. For examples of rubrics, visit www.irubrics.com . The schedule should be consistent with the academic calendar. Any significant changes should be avoided. If significant changes are made, academic leaders should be informed in writing, and students should be given a new schedule of assessment table).

Assessment	Assessment Task	Week Due	Proportion of Final Assessment
0	Attendance (0-4 absences: 5/5, then every absence gets -0.5)	-	5
1	Quiz 1	Week 4	5
2	Midterm	4/10/2026	15

		at 12:00 pm	
3	Project (part 1)	Week 9	10
5	Quiz 2	Week 13	5
6	Project (part 2)	Week 13	10
7	Lab exam	Week 11	10
8	Final Exam	finals	40

VII. Learning Resources

A. References:

a. Primary Textbook: S. McConnell, *"Code Complete: A Practical Handbook of Software Construction"*, Second Edition, Microsoft Press, ISBN-10: 0735619670, ISBN-13: 978-0735619678, 2nd Edition, 2004. (Used in Master Program)

b. Optional Text:

i. Working Effectively with Legacy Code, by Michael C. Feathers. Publisher: Pearson Education, Prentice-Hall ISBN-10: 0-13-117705-2

ii. Joshua Bloch. *Effective Java*. Second Edition. Addison Wesley, 2008, ISBN 0-321-35668-3

iii. Software Engineering (9th Edition) Ian Somerville

iv. Beautiful Code, Andy Oram & Greg Wilson, O'Reilly, 2007

v. Refactoring: Refactoring: Improving the Design of Existing Code, by Martin Fowler Publisher: Addison-Wesley Professional; 1 edition (July 8, 1999) ISBN-10: 0201485672

vi. Software Engineering Body of Knowledge (SWEBOK), http://www.computer.org/portal/web/swebok

c. Learning Management System: Moodle available at https://lms.psu.edu.sa

B. Facilities Required:

a. **Eclipse IDE for Enterprise Java and Web Developers:** https://www.eclipse.org/downloads/packages/release/2024-12/r/eclipse-ide-enterprise-java-and-web-developers

b. JDK 23, JavaFX 23

c. Google Chrome

C. Digital Tools:

None

D. Learning Management System – LMS using "Moodle" is required; items such as syllabus, course materials, assignments, rubrics, and announcements will be delivered through Moodle.

E. Online Courses: No online courses are scheduled. Only in cases on force majeure are courses delivered online through google meet.

F. Writing and Tutoring Center – Students are highly advised to use the Writing and Tutoring Center's academic services by booking tutoring/writing appointments through the **booking system:** (https://psu.mywconline.net/)

VIII. Classroom Policies

A. Academic Integrity Policy (e.g., plagiarism or dishonesty)

"Plagiarism can be defined as unintentionally or deliberately using AI-based tools or another person's writing or ideas as though they were one's own. Plagiarism includes, but is not limited to, using **ChaptGPT** or other AI tool to produce the project code or copying another individual's work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one's own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion and failure, or even suspension" (Academic Integrity and Syllabus Acknowledgement Form).

All students are expected to submit their 'own' work and not the work of others.

B. Attendance Policy

Please, adhere to the following guidelines:

1. The University attendance policy will be strictly followed. In this course, the absence of **(16)** hours results in a Denied Notice (DN). (Refer to the student regulation handbook for further details.)

Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a "DN" grade will be granted automatically during any time in the semester without any notice from the e-register system.

2. No make-up exams will be allowed except for university accepted documents. (Refer to the student regulation handbook for further details.)

C. Homework Submission Policy ***(****policy depends upon the course or the instructor****):***
Submissions that are less than one week LATE are graded out of 80% of the grade.
Submissions that are more than one week LATE are graded 0.

IX. My Assumptions

The student is expected to have these technical skills, otherwise he has the responsibility to learn them by himself before taking labs or entering any computer-based exam:

1. **Using Winrar or Winzip to zip and unzip multiple files or multiple folders.**
2. **Navigating a file system from a console:**

   (cd ./dir , cd .. , cd ../dir …)
3. **Programming:**
   a. **Conditionals (if..then..else)**
   b. **Loops (for, for each, while)**
   c. **Exception handling (try … catch)**
   d. **Objects: someObject.myFunction()**

The quality of the code the student writes is always taken into consideration in grading. Students have to apply the coding styles that are leaned throughout the course. For your reference, we are using simplified google code styles for Java (can be found on the LMS theory section) https://google.github.io/styleguide/javaguide.html